

Birzeit University
Faculty of Engineering & Technology
Computer Science Department
Encryption Theory
First Semester 2025/2026
Project Two

A group of students (three students per group) must select one of the two project areas to implement as their final project. Each group is required to complete all steps of the selected project carefully and to demonstrate clear understanding of the implementation details. In addition, each group must select a recent, high-quality research paper (recent work) that proposes an enhancement related to the chosen area, and implement the enhancement as part of the project.

A project discussion (oral evaluation) will be conducted, during which each group will be asked about the details of their design decisions, implementation steps, selected paper, evaluation metrics, and experimental results.

A scientific report is mandatory. The report must clearly present your work, justify your design choices, explain the implementation, and summarize the evaluation results. Using appropriate evaluation metrics is essential. Some evaluation techniques are listed in this document; however, you are encouraged to propose additional metrics or techniques to be creative and to achieve stronger results (this may improve your grade).

Deadline: 01/02/2026

Project discussion dates: 02/02/2026 – 03/02/2026

One group member must upload your complete work (report + code + results) to ITC under the Final Project title only. The deadline is strict, and there will be no postponement.

Stream Cipher Enhancement and Evaluation

Project Description

In this project, students are required to implement a conventional stream cipher and then propose and implement an enhanced version of the cipher. The goal of the enhancement is to improve the randomness and security properties of the generated keystream.

Students must compare the conventional and enhanced stream cipher designs using clearly defined evaluation metrics and demonstrate how the proposed enhancement performs under basic attack scenarios.

Report Requirements

Each group must submit a report of 8–10 pages, which must include:

1. Clear description of the selected stream cipher
2. Step-by-step explanation of the implementation
3. Description of the proposed enhancement
4. Experimental results and evaluation
5. Discussion of weaknesses and limitations
6. Analysis of resistance to basic attacks

Selected Topic: Stream Cipher Implementation and Enhancement

Each group must choose one conventional stream cipher as a baseline:

- Vernam cipher (XOR with keystream), **or**
- LFSR-based stream cipher

The generated keystream must be at least as long as the plaintext.

Part A: Baseline Stream Cipher Implementation

Students must:

1. Implement the selected conventional stream cipher.
2. Generate a keystream and encrypt a plaintext message.
3. Decrypt the ciphertext correctly to recover the original plaintext.
4. Use this implementation as the baseline solution for comparison.

Part B: Proposed Enhancements (Choose ONE or TWO)

Each of groups must enhance the baseline stream cipher by implementing one or two of the following techniques:

Option 1: Nonlinear Keystream Generation

- Introduce a simple nonlinear function in the keystream generation process.
- Compare the results with the conventional linear Vernam or LFSR-based cipher.

Option 2: Multiple LFSRs

- Use two LFSRs instead of one.
- Combine their outputs using a function such as XOR.

Option 3: Irregular Clocking

- Clock the LFSR irregularly (not at every step).
- Analyze how irregular clocking affects the keystream randomness.

Each enhancement must be clearly explained and justified.

Part C: Evaluation Metrics (Mandatory)

The baseline and enhanced stream ciphers must be evaluated using the following metrics:

Metric 1: Randomness (Bit Distribution)

- Count the number of 0s and 1s in the generated keystream.
- A good keystream should have nearly equal numbers of 0s and 1s.

Metric 2: Period Length

- Measure how long the keystream takes to repeat.
- A longer period indicates better security.

Metric 3: Avalanche Effect, you can use the following

- Flip one bit in the secret key.
- Generate a new keystream.
- Measure how many bits in the keystream change.
- Higher sensitivity indicates better resistance to attacks.

Part D: Attack Scenarios (Mandatory)

- Students must apply at least two of the following attacks:
- Keystream Reuse Attack
- Encrypt two different plaintexts using the same keystream.
- Analyze what information can be leaked.
- Known Plaintext Attack
- Assume the attacker knows part of the plaintext and ciphertext.
- Attempt to recover keystream bits and predict future output.

Students must compare how the **baseline** and **enhanced** ciphers behave under these attacks.

Comparative Analysis and Discussion

Each group must:

- Compare the baseline and enhanced ciphers using all evaluation metrics.
- Present results using tables and figures.
- Discuss whether the proposed enhancement improves security.
- Clearly state remaining weaknesses and limitations.

Important Notes

- Each enhancement must be implemented **independently** and compared with the baseline cipher.
- Copying real-world stream cipher designs is **not allowed**.
- Explanations must be written in **students' own words**.
- Code plagiarism will result in **zero marks**

Block Cipher Enhancement and Evaluation

study a conventional block cipher, implement an enhanced version proposed in a research paper (you must select one recent), and evaluate whether the enhancement improves security. Then each group must compare both designs experimentally and critique the enhanced solution using fundamental block cipher security principles, particularly Shannon's principles of confusion and diffusion.

Each group must submit a report of 8–10 pages, which must include:

1. Description of the selected block cipher
2. Summary of the selected research paper (which means main contribution for this research).
3. Step-by-step implementation details (try to change the parameters to enhance the security of proposed solution).
4. Experimental evaluation and results
5. Security critique based on block cipher principles.
6. Discussion of strengths, weaknesses, and limitations

Each group must choose ONE conventional block cipher as a baseline:

- Feistel cipher (custom or textbook version), **or**
- DES algorithm.
- Mini-AES
- Hill-cipher

Each group must select ONE research paper that proposes an improvement to the chosen block cipher.

Examples of acceptable enhancements include:

- Increasing diffusion
- Modifying round functions
- Improving key scheduling
- Increasing resistance to known attacks

Regarding to evaluation, each group must evaluate both baseline (conventional) and enhanced block cipher as follows:

You must check if algorithm achieve which level of avalanche effect. As example (you must check for other procedure to show affect) you can flip one bit in plain text and encrypt again using same key, then measure how many bits change in the ciphertext. Secure solution must show close to 50%-bit change.

Second metric of evaluation refers to key distribution, you must be creative, same idea can be used as example (search for another metric for this purpose) that flip one bit in the key then show how the generated cipher text change.

Third metric which refers to evaluate diffusion, then the evaluation of encryption output patterns.

Then each group must build some comparative result that represent results using tables and figures, that compare baseline and enhanced designs clearly. Explain which design performs better and why.

Each group must evaluate both conventional and enhanced solution from Known Limitations. Can the cipher still be vulnerable to: Brute-force attack? Known-plaintext attack? Differential-style attacks (conceptual discussion only)?